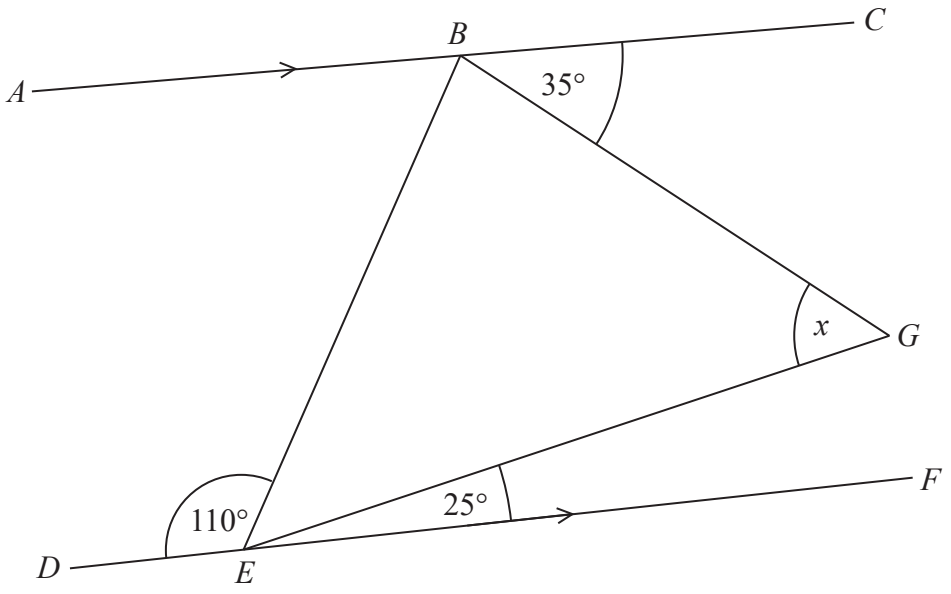
**3** *BEG* is a triangle.




*ABC* and *DEF* are parallel lines.

Work out the size of angle *x*.
Give a reason for each stage of your working.

ABE = 180° – 110° = 70°

Co-interior angles on parallel lines add to 180°.

**1 mark**
Use the parallel lines

EBG = 180° – 70° – 35° = 75°

Angles on a straight line add to 180°.

BEG = 180° – 110° – 25° = 45°

Angles on a straight line add to 180°.

x = 180° – 75° – 45° = 60°

Angles in a triangle add to 180°.

**1 mark**
Complete the angle calculation

**1 mark**
Link correct parallel-line and triangle/straight-line reasons

**1 mark**
Correct angle

60 °

**(Total for Question 3 is 4 marks)**